

ONE-SIDED



Limits



LEFT-HAND AND RIGHT-HAND LIMITS

- Let the function f be defined on the interval (x, c) such that $c > x$. If f approaches the number L as x approaches c within (x, c) , then L is called the left hand limit of f at c .

$$\lim_{x \rightarrow c^-} f(x) = L$$

LEFT-HAND AND RIGHT-HAND LIMITS

- Similarly, if f is defined on the open interval (c, x) such that $c < x$, and it approaches the number M as x approaches c within (c, x) , then M is called right-hand limit of f at c .

$$\lim_{x \rightarrow c^+} f(x) = M$$

LEFT-HAND AND RIGHT-HAND LIMITS

1. Consider the function
 $f(x) = \sqrt{4x - 2}$, $x \geq \frac{1}{2}$.
Evaluate $\lim_{x \rightarrow \frac{1}{2}^+} f(x)$.

SOLUTION:

- The expression $\lim_{x \rightarrow \frac{1}{2}^+} f(x)$ represents the limit of the function as x approaches $\frac{1}{2}$ or 0.5 from the right. Thus, the limit can be calculated by assigning values greater than but arbitrarily close to 0.5. Using the calculator, you can build the following table of values:

SOLUTION: ||

- Table 2.1. Table of values for example 1

- As shown in table 2.1, $\lim_{x \rightarrow \frac{1}{2}^+} f(x) = 0$.

LEFT-HAND AND RIGHT-HAND LIMITS

2. Consider the following function:

$$h(x) = \begin{cases} x^2 + 4x, & x \leq 2 \\ \frac{3}{2}x + 9, & x > 2 \end{cases}$$

Evaluate $\lim h(x)$

SOLUTION:

- The value of $\lim_{x \rightarrow 2} h(x)$ will only exist if $\lim_{x \rightarrow 2^-} h(x) = \lim_{x \rightarrow 2^+} h(x)$. To get $\lim_{x \rightarrow 2^-} h(x)$, substitute $x = 2$ on the first equation $(x^2 + 4x)$. By direct substitution, $\lim_{x \rightarrow 2^-} h(x) = 2^2 + 4(2) = 12$. To get $\lim_{x \rightarrow 2^+} h(x)$, substitute $x = 2$ in the second equation $(\frac{3}{2}x + 9)$. Therefore, $\lim_{x \rightarrow 2^+} h(x) = \frac{3}{2}(2) + 9 = 12$. Thus, $\lim_{x \rightarrow 2} h(x) = 12$.

LEFT-HAND AND RIGHT-HAND LIMITS

3. Consider the following function:

$$f(x) = \begin{cases} 2A - Bx, & x > 1 \\ 4, & x = 1 \\ A + Bx^2, & x < 1 \end{cases}$$

Find the values of A and B so that $\lim_{x \rightarrow 1} f(x)$ exist.

SOLUTION:

- For the limit exist, the left-hand and the right-hand limits must be equal to 4 because $f(1) = 4$.

$$\lim_{x \rightarrow 1^-} f(x) = \lim_{x \rightarrow 1^+} f(x)$$

SOLUTION:

- By direct substitution,

$$\lim_{x \rightarrow 1^-} f(x) = 4 \Leftrightarrow A + B(1)^2 = 4 \Leftrightarrow A + B = 4$$

$$\lim_{x \rightarrow 1^+} f(x) = 4 \Leftrightarrow 2A - B(1) = 4 \Leftrightarrow 2A - B = 4$$

- Thus, you have the system of equations:
$$\begin{cases} A + B = 4 \\ 2A - B = 4 \end{cases}$$

SOLUTION:

- By solving for A using addition method, you have

$$\begin{array}{r} A + B = 4 \\ + 2A - B = 4 \\ \hline 3A = 8 \\ 8 \\ A = \frac{8}{3} \end{array}$$

SOLUTION:

- Solving for B using $A + B = 4$, you have $B = 4 - A$:

$$B = 4 - \frac{8}{3}$$

$$B = \frac{4}{3}$$

WHAT HAVE I LEARNED SO FAR?

- Let

$$f(x) = \begin{cases} x + 2, & x > 0 \\ -x - 6, & x < 0 \end{cases}$$

WHAT HAVE I LEARNED SO FAR?

- a. Find $\lim_{x \rightarrow 0^+} f(x)$ and $\lim_{x \rightarrow 0^-} f(x)$.
- b. Does $\lim_{x \rightarrow 0} f(x)$ exist? If yes, give its value. If not, why?
- c. Find $\lim_{x \rightarrow -4^+} f(x)$ and $\lim_{x \rightarrow -4^-} f(x)$.
- d. Does $\lim_{x \rightarrow -4} f(x)$ exist? If yes, give its value. If not, why?

INFINITE



Limits



INFINITE LIMITS

- If the value of $f(x)$ arbitrarily increases positively for all x sufficiently close to c from both sides, without letting x equal to c , then

$$\lim_{x \rightarrow c} f(x) = \infty$$

INFINITE LIMITS

- *Likewise*, If the value of $f(x)$ arbitrarily increases negatively for all x sufficiently close to c from both sides, without letting x equal to c , then

$$\lim_{x \rightarrow c} f(x) = -\infty$$

INFINITE LIMITS

Evaluate the following limit:

$$\lim_{x \rightarrow 0} \frac{4}{x^3}$$

SOLUTION:

- By direct substitution, you write $\lim_{x \rightarrow 0} \frac{4}{x^3} = \frac{0}{0}$, which makes it undefined. This is a hint that the function approaches infinity. You just need to reconcile whether it is a positive or a negative infinity. To get the limit from the left of, x should be as close to 0 as possible, so you use -0.00001 . Thus, you

SOLUTION:

- $\frac{4}{x^3} = \frac{4}{(-0.00001)^3} = -4 \times 10^{15}$
- This means that from the left, the graph approaches $-\infty$ as x approaches 0; so, $\lim_{x \rightarrow 0^-} \frac{4}{x^3} = -\infty$.

SOLUTION:

- To get the limit from the right, let $x = 0.00001$. Then,

$$\frac{4}{x^3} = \frac{4}{(0.00001)^3} = 4 \times 10^{15}$$

- Therefore, $\lim_{x \rightarrow 0^+} \frac{4}{x^3} = \infty$. Finally, because

$$\lim_{x \rightarrow 0^-} \frac{4}{x^3} = -\infty \neq \lim_{x \rightarrow 0^+} \frac{4}{x^3} = \infty, \text{ then } \lim_{x \rightarrow 0} \frac{4}{x^3} \text{ does not exist.}$$

INFINITE LIMITS

Evaluate the following limit:

a.

$$\lim_{x \rightarrow -1^-} \frac{-6x^2}{x^2 - 1}$$

SOLUTION:

- By direct substitution, you have

$$\lim_{x \rightarrow -1^-} \frac{-6x^2}{x^2 - 1} = \frac{-6(-1)^2}{(-1)^2 - 1} = \frac{-6}{0}$$

- Thus you have infinite limit.

SOLUTION:

a) To get the specified limit, let x be sufficiently close to -1 from the left and substitute it to a given function. Use $x = -1.0001$.

$$\frac{-6x^2}{x^2 - 1} = \frac{-6(-1.0001)^2}{(-1.0001)^2 - 1} = -30\,004.5$$

• Thus, $\lim_{x \rightarrow -1^-} \frac{-6x^2}{x^2 - 1} = -\infty$

INFINITE LIMITS

Evaluate the following limit:

b.

$$\lim_{x \rightarrow -1^+} \frac{-6x^2}{x^3}$$

SOLUTION:

b) Let x be equal to -0.9999 . Substituting in the function,

$$\frac{-6x^2}{x^2 - 1} = \frac{-6(-0.9999)^2}{(-0.9999)^2 - 1} = 29\,995.5$$

• Thus, $\lim_{x \rightarrow -1^+} \frac{-6x^2}{x^2 - 1} = \infty$

INFINITE LIMITS

Evaluate the following limit:

c.

$$\lim_{x \rightarrow -1} \frac{-6x^2}{x^3}$$

SOLUTION:

• c) Because $\lim_{x \rightarrow -1^-} \frac{-6x^2}{x^2 - 1} \neq \lim_{x \rightarrow -1^+} \frac{-6x^2}{x^2 - 1}$, then
 $\lim_{x \rightarrow -1} \frac{-6x^2}{x^3}$ does not exist.

LIMITS



at Infinity



LIMITS AT INFINITY

- $$f(x) = \frac{1}{x}$$
- As x increases infinity without bound
- As x approaches infinity on the negative side.

THE LIMIT OF $\frac{1}{x}$ AT INFINITY

- As x approaches infinity from both sides, the value of $f(x) = \frac{1}{x}$ approaches 0. In symbols, you have

$$\lim_{x \rightarrow \pm \infty} \frac{1}{x} = 0$$

THE LIMIT OF $\frac{1}{r}$ AT INFINITY

- Evaluate $\lim_{x \rightarrow \infty} \frac{3}{x}$
- Evaluate $\lim_{x \rightarrow -\infty} \frac{5}{x^2}$

SOLUTION:

- Apply the limit laws

$$\begin{aligned}\lim_{x \rightarrow \infty} \frac{3}{x} &= \lim_{x \rightarrow \infty} \left(3 \cdot \frac{1}{x} \right) \\ &= 3 \cdot \lim_{x \rightarrow \infty} \frac{1}{x} \\ &= 3 \cdot 0 \\ &= 0\end{aligned}$$

SOLUTION:

- Apply the limit laws

$$\begin{aligned}\lim_{x \rightarrow -\infty} \frac{5}{x^2} &= \lim_{x \rightarrow -\infty} \left(5 \cdot \frac{1}{x^2} \right) \\ &= 5 \cdot \lim_{x \rightarrow -\infty} \frac{1}{x} \cdot \lim_{x \rightarrow -\infty} \frac{1}{x} \\ &= 5 \cdot 0 \cdot 0 \\ &= 0\end{aligned}$$

THE LIMIT OF $\frac{k}{x^n}$ AT INFINITY

- For all real numbers k and a positive rational numbers n ,

$$\lim_{x \rightarrow \pm\infty} \frac{k}{x^n} = 0$$

LIMITS AT INFINITY

- Evaluate $\lim_{x \rightarrow \infty} \frac{x^2 - 2x - 3}{x^2 - 4}$
- Evaluate $\lim_{x \rightarrow -\infty} \frac{x^3 - 12}{4x^3 + 5}$

SOLUTION:

- To simplify the process of evaluating limits at infinity, you should always get a term of the form $\frac{k}{x^n}$. To do this, you need to divide each term of the function by the highest power which is x^2 . Thus,

SOLUTION:

- $$\lim_{x \rightarrow \infty} \frac{x^2 - 2x - 3}{x^2 - 4} = \lim_{x \rightarrow \infty} \frac{\frac{x^2}{x^2} - \frac{2x}{x^2} - \frac{3}{x^2}}{\frac{x^2}{x^2} - \frac{4}{x^2}}$$
$$= \lim_{x \rightarrow \infty} \frac{\left(1 - \frac{2}{x} - \frac{3}{x^2}\right)}{\left(1 - \frac{4}{x^2}\right)}$$

SOLUTION: 

$$\begin{aligned} & \bullet \\ &= \frac{\lim_{x \rightarrow \infty} 1 - \lim_{x \rightarrow \infty} \frac{2}{x} - \lim_{x \rightarrow \infty} \frac{3}{x^2}}{\lim_{x \rightarrow \infty} 1 - \lim_{x \rightarrow \infty} \frac{4}{x^2}} \\ &= \frac{1 - 0 - 0}{1 - 0} \\ &= 1 \end{aligned}$$

SOLUTION:

•

$$\begin{aligned}\lim_{x \rightarrow -\infty} \frac{x^3 - 12}{4x^3 + 5} &= \lim_{x \rightarrow -\infty} \frac{\frac{x^3}{x^3} - \frac{12}{x^3}}{\frac{4x^3}{x^3} + \frac{5}{x^3}} \\ &= \lim_{x \rightarrow -\infty} \frac{\left(1 - \frac{12}{x^3}\right)}{\left(4 + \frac{5}{x^3}\right)}\end{aligned}$$

SOLUTION: 

•

$$= \frac{\lim_{x \rightarrow -\infty} 1 - \lim_{x \rightarrow -\infty} \frac{12}{x^3}}{\lim_{x \rightarrow -\infty} 4 + \lim_{x \rightarrow -\infty} \frac{5}{x^3}}$$
$$= \frac{1 - 0}{4 + 0} = \frac{1}{4}$$

LIMITS AT INFINITY

• Evaluate $\lim_{x \rightarrow \infty} \frac{4x^2 - 3}{x + 4}$

SOLUTION:

- $$\begin{aligned}\lim_{x \rightarrow \infty} \frac{4x^2 - 3}{x + 4} &= \lim_{x \rightarrow \infty} \frac{\frac{4x^2}{x^2} - \frac{3}{x^2}}{\frac{x}{x^2} + \frac{4}{x^2}} \\ &= \lim_{x \rightarrow \infty} \frac{\left(4 - \frac{3}{x^2}\right)}{\left(\frac{1}{x} + \frac{4}{x^2}\right)}\end{aligned}$$

SOLUTION:

- $$= \frac{\lim_{x \rightarrow \infty} 4 - \lim_{x \rightarrow \infty} \frac{3}{x^2}}{\lim_{x \rightarrow \infty} \frac{1}{x} + \lim_{x \rightarrow \infty} \frac{4}{x^2}}$$

$$= \frac{4 - 0}{0 + 0} = \frac{4}{0}$$

- Because the value of $\frac{4}{0}$ is undefined, the limit reaches infinity (∞).

WHAT HAVE I LEARNED SO FAR?

- Evaluate $\lim_{x \rightarrow -\infty} \frac{5-4x-6x^3}{x^2-2}$
- Evaluate $\lim_{x \rightarrow \infty} (3x^2 - 2x)$
- Evaluate $\lim_{x \rightarrow -\infty} \sqrt{\frac{4x^2+2x-1}{9x^2-x-3}}$